Date: March 17, 2026
To: Prof Nate Sniadeci, Vinh Nguyen, Zachary Lewandowski
From: Kit Nguyen
Subject: Analysis of Climbing Structure

EXECUTIVE SUMMARY:
The playground structure needs to be analysed to determine if it is safe for children to play on. The trusses were created with 11 nodes and 25 elements, with the bottom 5 nodes being either fixed or free supports. Analysing the playground structure for fixed support vs free support under at 5000N force, it was able to be determined that using the larger diameter (4cm) steel trusses results in less deformation than than the smaller diameter ones (2cm). Deformation and axial forces between fixed and non-fixed support for similar size diameters show minimal changes.

INTRODUCTION:
The playground structure for the City of Seattle at Gasworks Park that is analyzed is shown below in **Figure 1.**

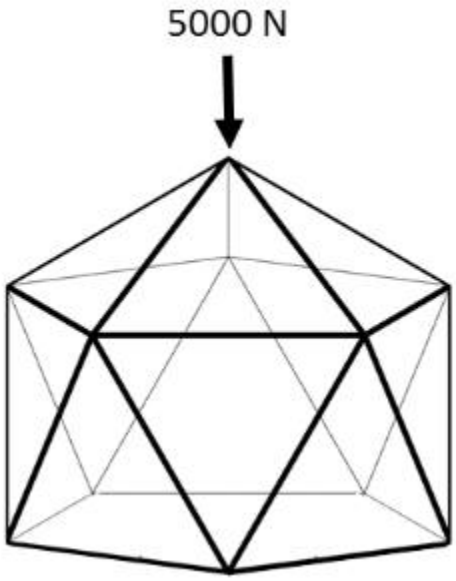


Figure 1: Playground climbing structure with load applied.

The purpose of this analysis is to determine if the properties of the steel truss (**Table 1**) are strong enough to support a worse-case scenario of 5000N acting at the top of the structure. The ANSYS model was created to model different size steel rods with circular cross sections. The rods were either 4cm (Case 1 and 3) or 2cm (Case 2 and 4) in diameter. The bottom 5 nodes of the playground structure either represented if the structure was staked (fixed boundary condition) to the ground (Case 1 and 2) or just sitting on top (free displacement boundary condition) of the ground (Case 3 and 4).

Table 1: Properties of the Steel

Elastic Modulus (GPa)	200
Poisson's Ratio	0.32

METHODS

The 3D model of the truss was created in ANSYS using the Ansys 2024 R2 Design Modeler. Starting with drawing singular nodes in 2D then connecting them with truss elements in 3D. Then a cross section was applied to each of the elements, allowing the trusses to be easily manipulated for different cross sections. The node locations are shown in **Table 2.**

Table 2: Playground structure node locations.

Node	X	Y	Z
1	0.851	0.000	0.000
2	0.263	0.809	0.000
3	-0.688	0.500	0.000
4	-0.688	-0.500	0.000
5	0.263	-0.809	0.000
6	0.688	0.500	0.688
7	-0.263	0.809	0.688
8	-0.851	0.000	0.688
9	-0.263	-0.809	0.688
10	0.688	-0.500	0.688
11	0.000	0.000	1.214

There were 4 different cases that were tested in ANSYS. Each of the cases were for circular cross sections of the truss.

1. Truss members have a diameter of 4 cm and nodes 1-5 have fixed boundary conditions.
2. Truss members have a diameter of 2 cm and nodes 1-5 have fixed boundary conditions.
3. Truss members have a diameter of 4 cm and nodes 1-5 have free boundary conditions.
4. Truss members have a diameter of 2 cm and nodes 1-5 have free boundary conditions.

The images for the setup for these are shown in **Appendix 1**. For each of the cases, a 5000N force was applied to the top central node of the structure.

RESULTS:

The results of the analysis are shown in **Table 3**.

Table 3: Analysis of playground structure

Column 1	Case 1: D = 4 cm, fixed	Case 2: D = 2 cm, fixed	Case 3: D = 4 cm, free	Case 4: D = 2 cm, free
Max Total Deformation (m)	2.70E-05	1.08E-04	2.70E-05	1.09E-04
Max Axial Compression (N)	-1.90E+03	-1.90E+03	-1.89E+03	-1.89E+03
Max Axial Tension (N)	1.58E+03	1.57E+03	1.56E+03	1.57E+03
Max Compressive Stress (Pa)	1.51E+06	6.05E+06	1.51E+06	6.01E+06
Max Tensile Stress (Pa)	1.25E+06	5.01E+06	1.24E+06	5.01E+06
Max Compressive Strain	7.57E-06	3.02E-05	7.53E-06	3.01E-05
Max Tensile Strain	6.27E-06	2.50E-05	6.22E-06	2.50E-05
Critical Buckling Load (N)	2.48E+05	1.55E+04	2.48E+05	1.55E+04
FOS Stress	1.85E+02	4.63E+01	1.86E+02	4.66E+01
FOS Buckling	1.30E+02	8.16E+00	1.31E+02	8.20E+00

Cases 1 and 3 have fixed boundary conditions on the ground nodes which represent a structure that is staked to the ground, so there is no movement in the X, Y, or Z direction. Case 2 and 4 have free boundary conditions on the ground nodes, which represent a structure that is just placed on the ground, so there is no movement in the Z direction, but movement in the X and Y directions.

The trusses in this analysis are pinned together, which represents it being bolted together. These joints allow for rotation at the connection points, whereas if it were welded, it would be rigid at these areas and have no movement. Also, these elements only transmit axial forces and no moment or bending forces, which is a characteristic of bolted joints.

Analysing displacement and loads for all 4 cases, it is seen that the largest displacement was always at the top node (Node 11). This makes sense because this is where the load is, and it is

also the furthest from the gound, so it has the most freedom to move and be displaced (see **Appendix 2)**. This also resulted in the highest compressive load to happen in the angled top trusses that connect with node 11 since they are taking the 5000N load (these are shown as the blue truss members at the top). The tensile load is greatest in the horizontal truss members that are in the middle ring that is directly supporting the compressive load of the angled truss members. This is because this ring is meant to resist the compressive load that is pushing on it radially, so it is stretching (in tension).

Using these ANSYS results, the stress and strain of each node was able to be found by using the the following equations:

$$\sigma = F/A$$
$$\epsilon = \sigma/E$$

Looking at **Table 2**, the 4 cm (Case 1 and 3) and 2 cm (Case 2 and 4) cross section cases can be compared. Stresses in 2 cm cross section trusses were typically 4x higher than 4cm, and this is because the axial forces were typically very similar (only ~10N difference between maximum and minimum axial forces between compressive and tension cases). As a result, the only thing that changes is the area, so by decreasing the diameter by 2 cm, the area decreases by 4 and you get an inverse relationship between cross sectional area and stress of the truss elements. This directly impacts that strain as well because the increase in the amount of stress is proportional to the increase in strain since the elastic modulus of steel is the same. This means strain in the 2 cm cases is also roughly 4x larger than in the 2 cm cases.

The maximum stress that occurred for the structure happened in the maximum compressive force locations. For the 4 cm cross sectional area cases, this was about 1.51 MPa (for both case 1 and 3) and roughly 6.05 MPa (for cases 2 and 4). When compared to the maximum yield strength of steel members, which is 280 - 1600 MPa, it can be concluded that the playground structure is safe from failure under these conditions regardless of which steel rod size is used by a factor of safety of 46 for worst case scenarios (2 cm cross section and lowest yield strength).

The same analysis can be done to see if the structure is as risk for buckling using Euler's equation for buckling:

$$P_{cr} = \pi^2 EI/L^2$$
$$I = \pi D^4/64$$

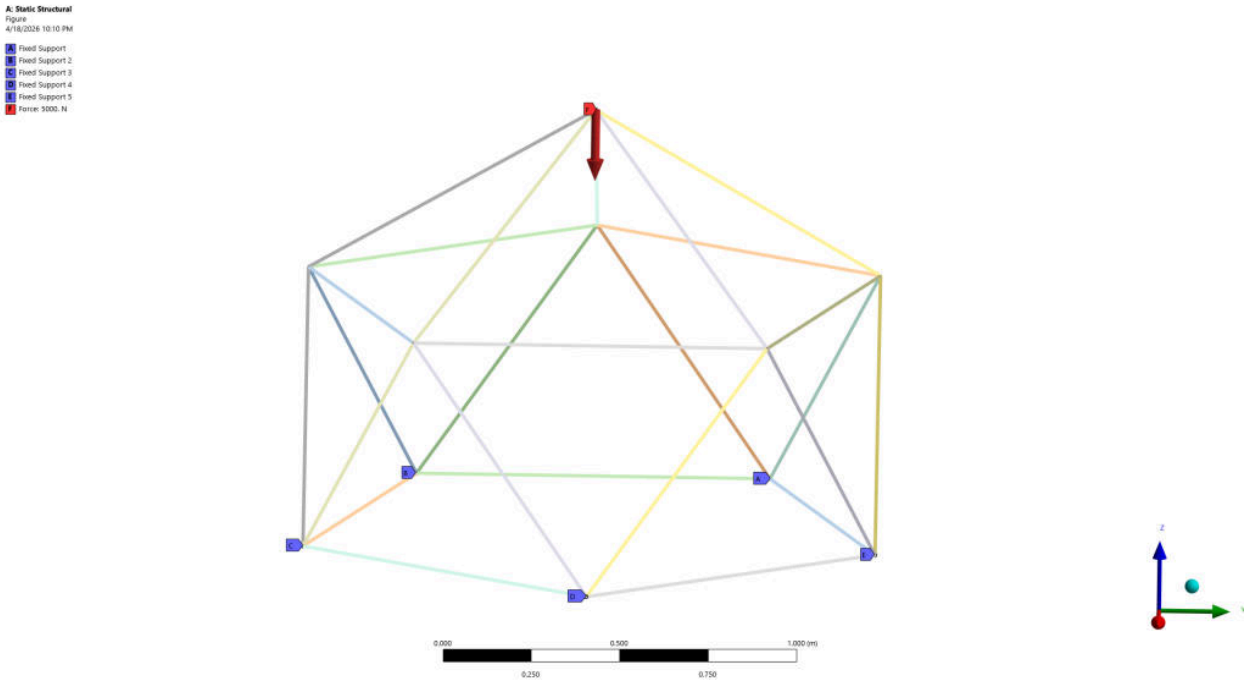
Using L = 1m (length of element where the axial load is the highest), we find that the lowest critical buckling load is, again, in the 2cm cross section cases. Even with this being lower, it is still seen that under these worse case scenarios, the failure due to buckling is unlikely to happen (factor of safety of 8.16).

CONCLUSION:

These analyses show that the playground structure is safe to operate under both cross sectional areas. Under worst case scenarios with a 5000 N downwards force, the FOS for failure under stress and buckling is 46 and 8.16 respectively. However, it is still recommended that the larger cross sectional area steel is used to allow for maximum safety on the play structure. Although the analysis is complete, errors could have occurred in this analysis because it doesn't account for specific scenarios, such as loading on multiple locations at once or loading at angles which the steel rod may not support as well (such as diagonally i.e. if someone was swinging on it). Therefore, future analysis could look at these scenarios to further validate this playground structure.

APPENDIX 1

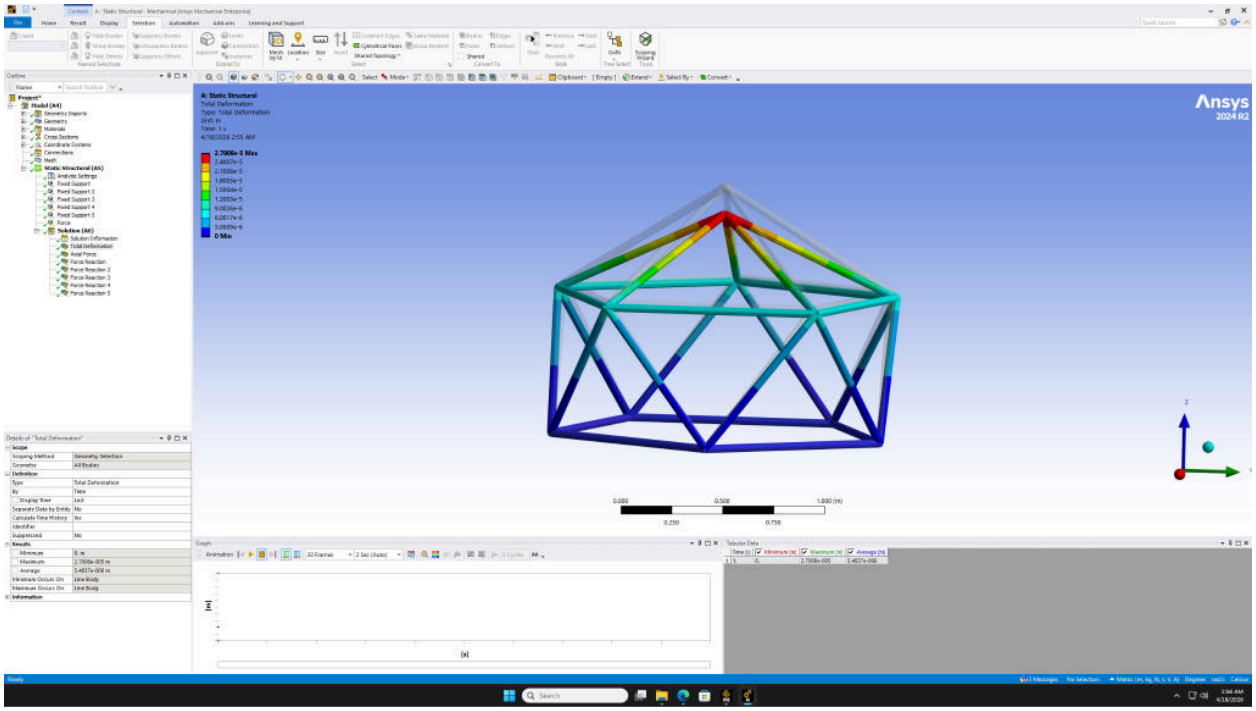
Boundary conditions for the tested FEA cases



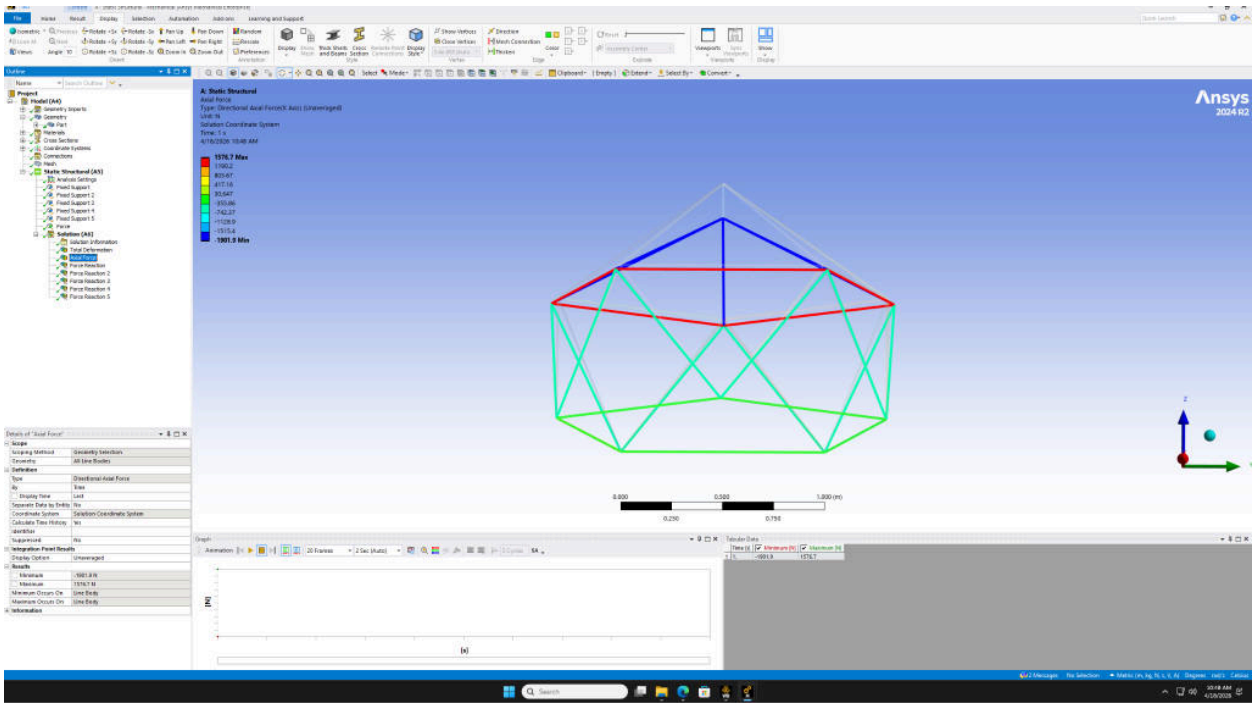
Appendix Figure 1a: Fixed Boundary

APPENDIX 2

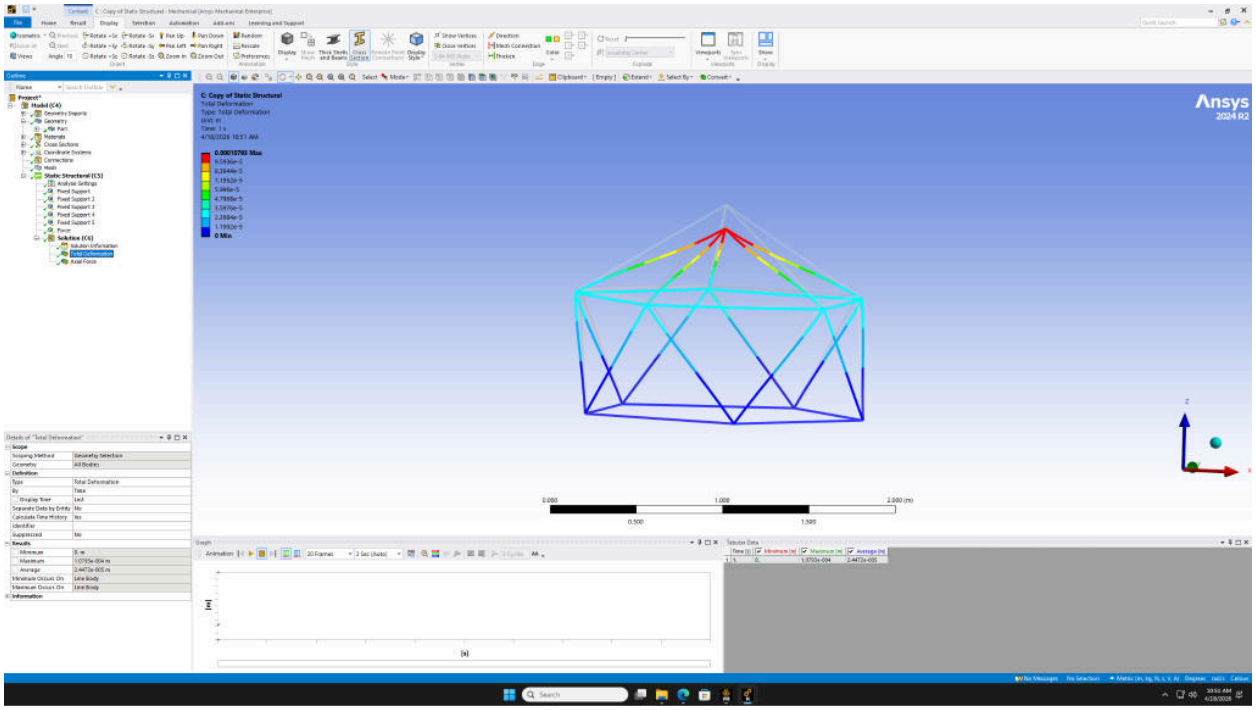
Force and Displacement of playground structure



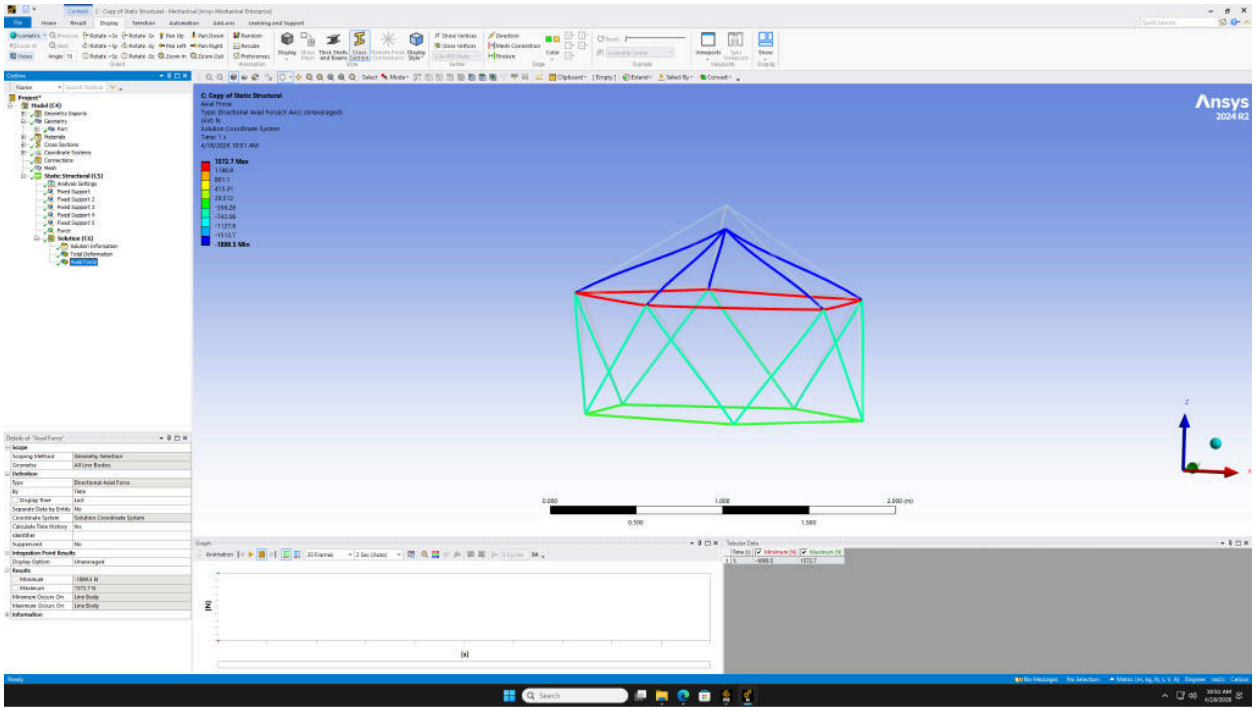
Appendix Figure 2a: Total deformation of case 1



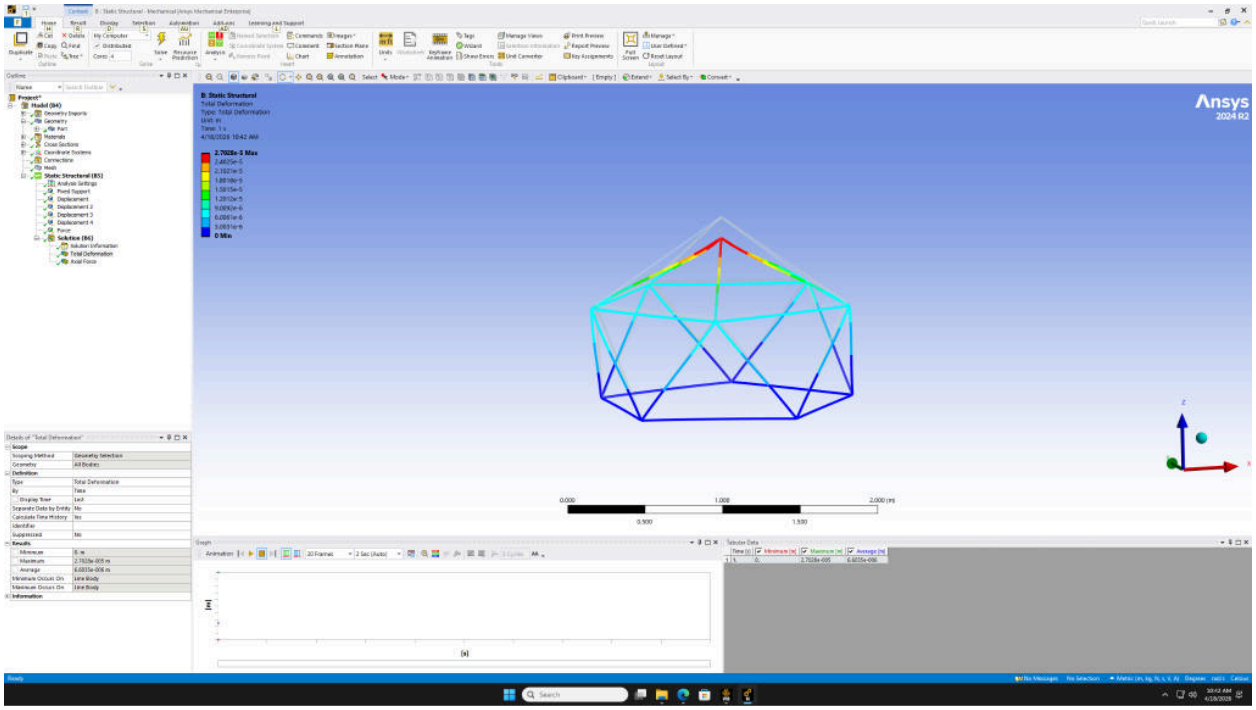
Appendix Figure 2b: Axial force of case 1



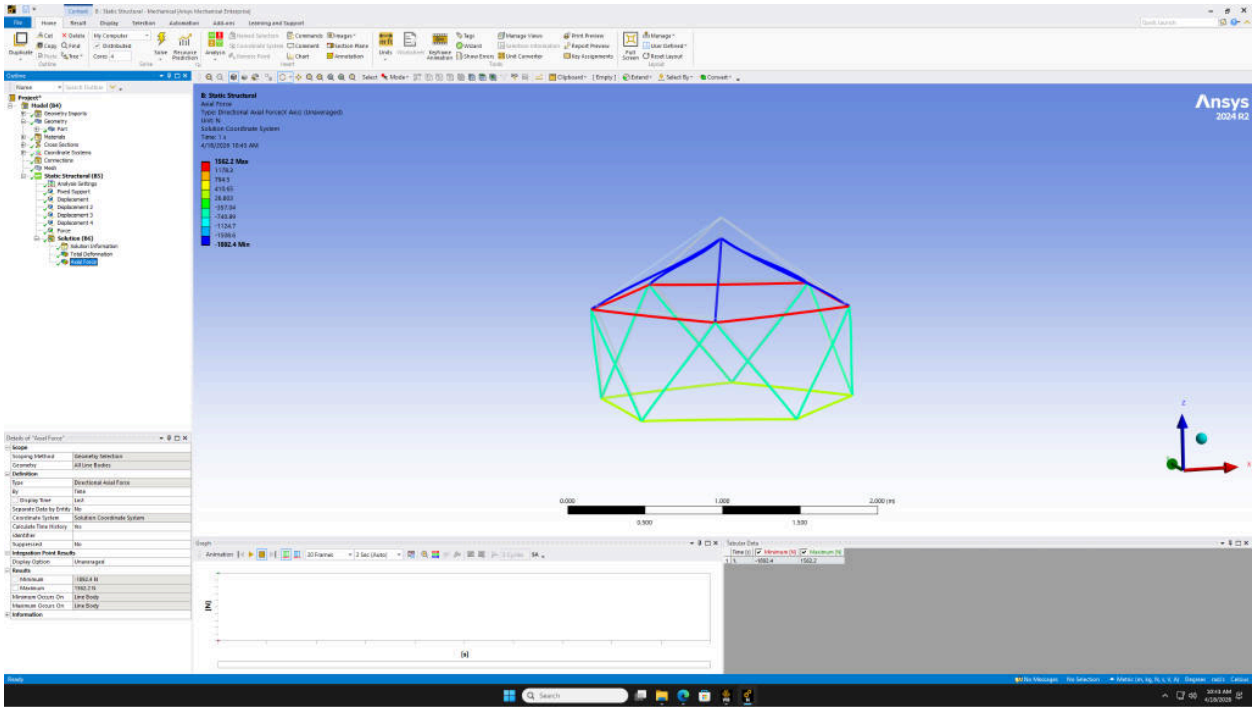
Appendix Figure 2c: Total deformation of case 2



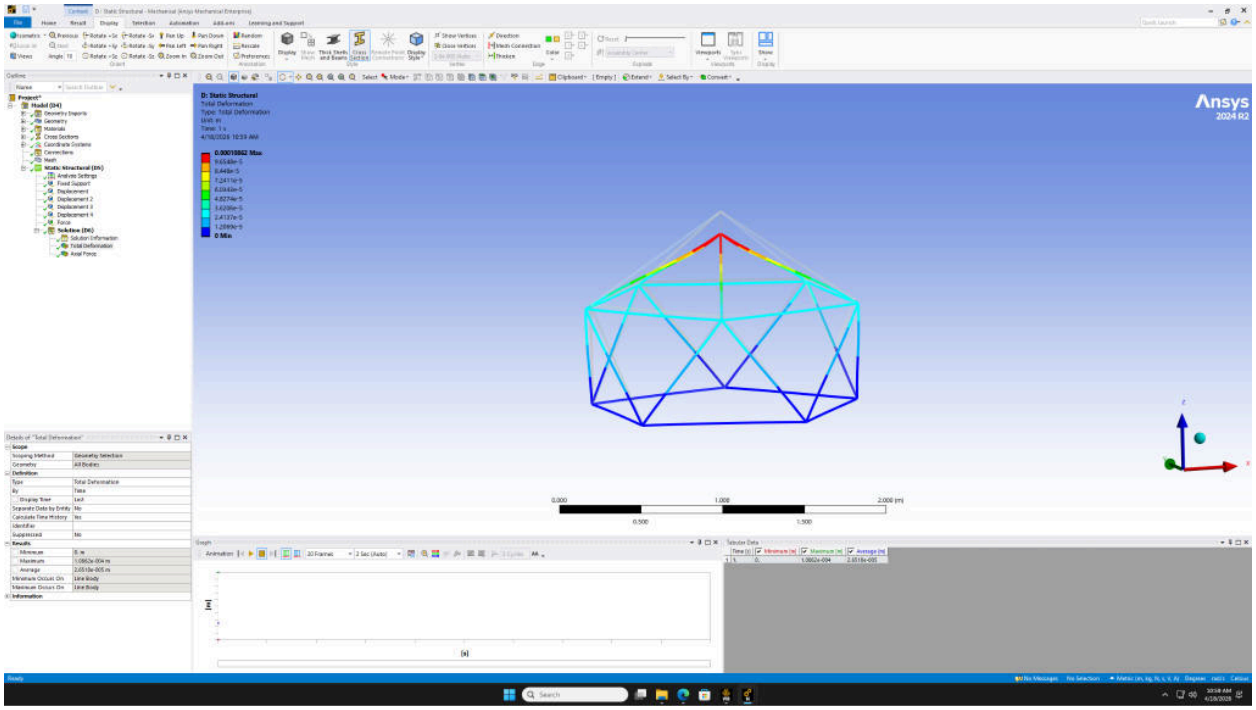
Appendix Figure 2d: Axial force of case 2



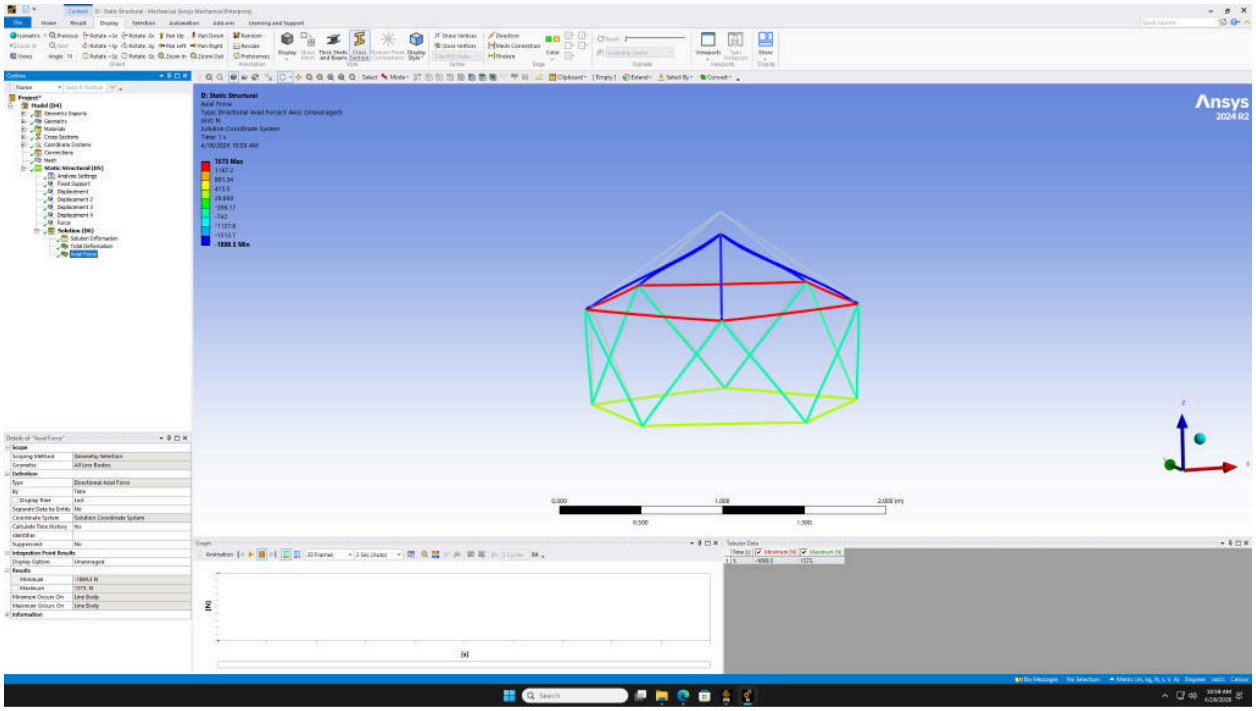
Appendix Figure 2e: Total deformation of case 3



Appendix Figure 2f: Axial force of case 3



Appendix Figure 2g: Total deformation of case 4



Appendix Figure 2h: Axial force of case 4

APPENDIX 3
DATA

Case 1: 4cm diameter, fixed

○ Available Solution Quantities

◉ Material and Element Type Information

○ Solver Component Names

○ Result Summary

Material IDs	Element Name IDs	Element Type IDs	Number of Elements
MAT_1-25	LINK180	1-25	25
MAT_26	FOLLW201	26	1
MAT_27	FOLLW201	26	1
MAT_28	FOLLW201	26	1

Element Name IDs	Material IDs	Element Type IDs	Number of Elements
LINK180	1-25	1-25	25
FOLLW201	26-28	26	3

Element Type IDs	Element Shape	Material IDs	Element Name IDs	Number of Elements
ETYPE_1-25	BEAM3	1-25	LINK180	25
ETYPE_26	POINT	26, 27, 28,	FOLLW201,	3

○ Available Solution Quantities

○ Material and Element Type Information

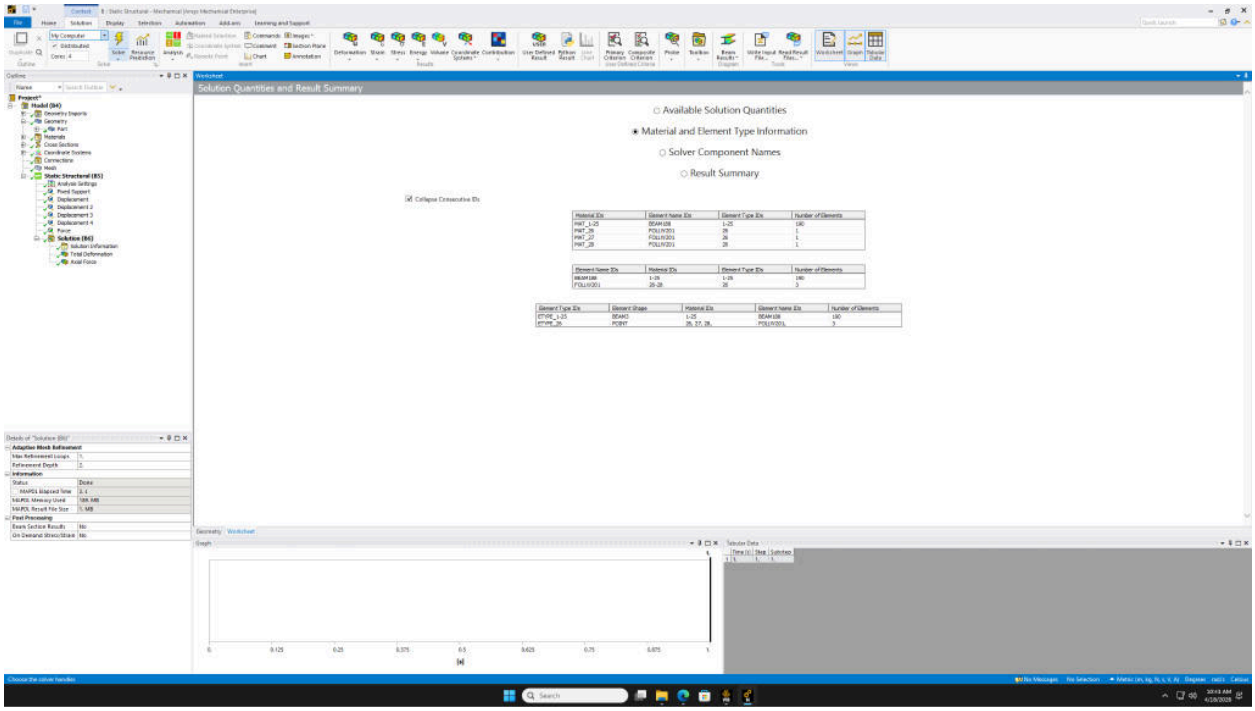
○ Solver Component Names

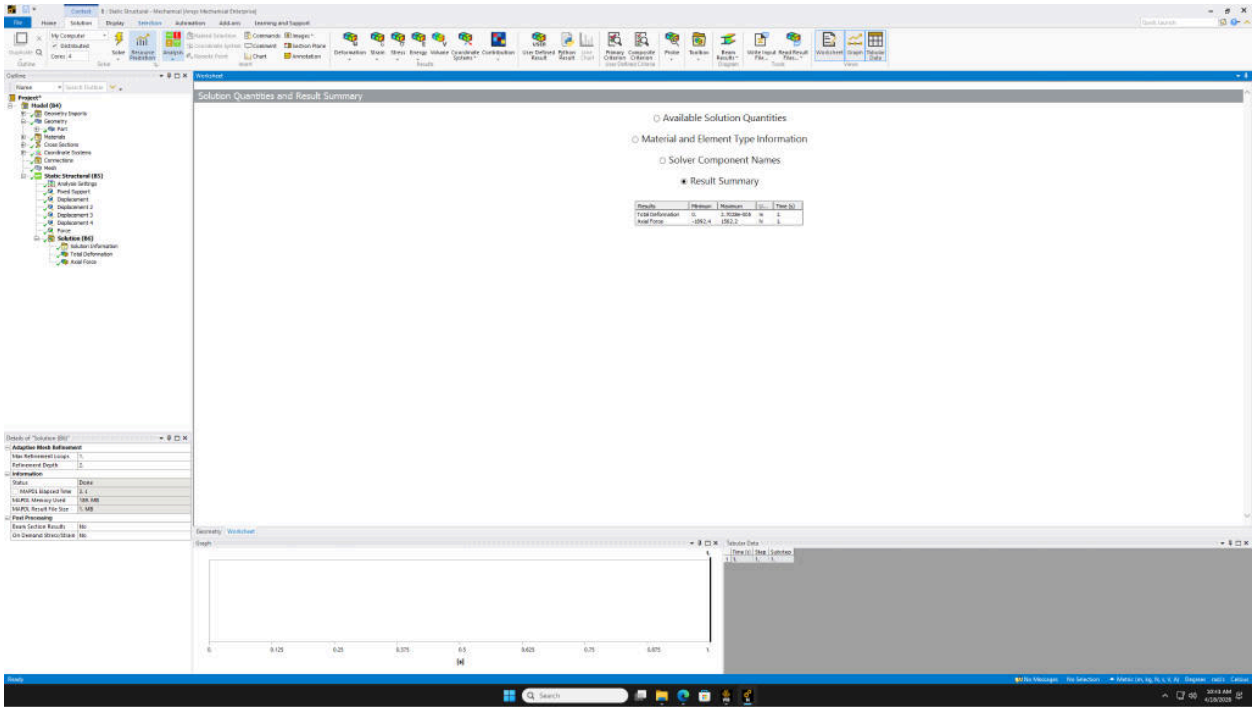
◉ Result Summary

Results	Minimum	Maximum	U...	Time (s)
Total Deformation	0.	2.7008e-005	m	1.
Axial Force	-1901.9	1576.7	N	1.

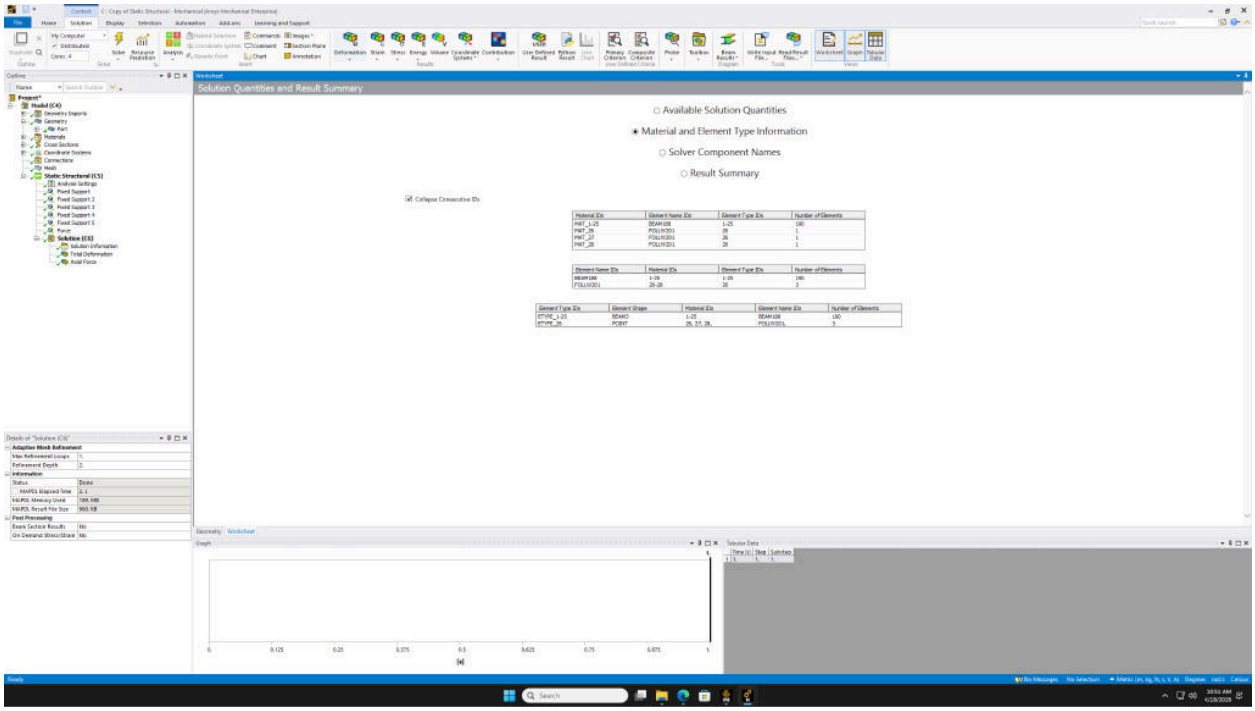
Probe: Reactions	X Magnitude	Y Magnitude	Z Magnitude	Total	U...	Time (s)
Force Reaction	190.81	-138.39	1000.3	1027.7	N	1.
Force Reaction 2	190.81	138.39	1000.3	1027.7	N	1.
Force Reaction 3	-72.418	224.87	999.94	1027.5	N	1.
Force Reaction 4	-236.78	-4.5475e-013	999.42	1027.1	N	1.
Force Reaction 5	-72.418	-224.87	999.94	1027.5	N	1.

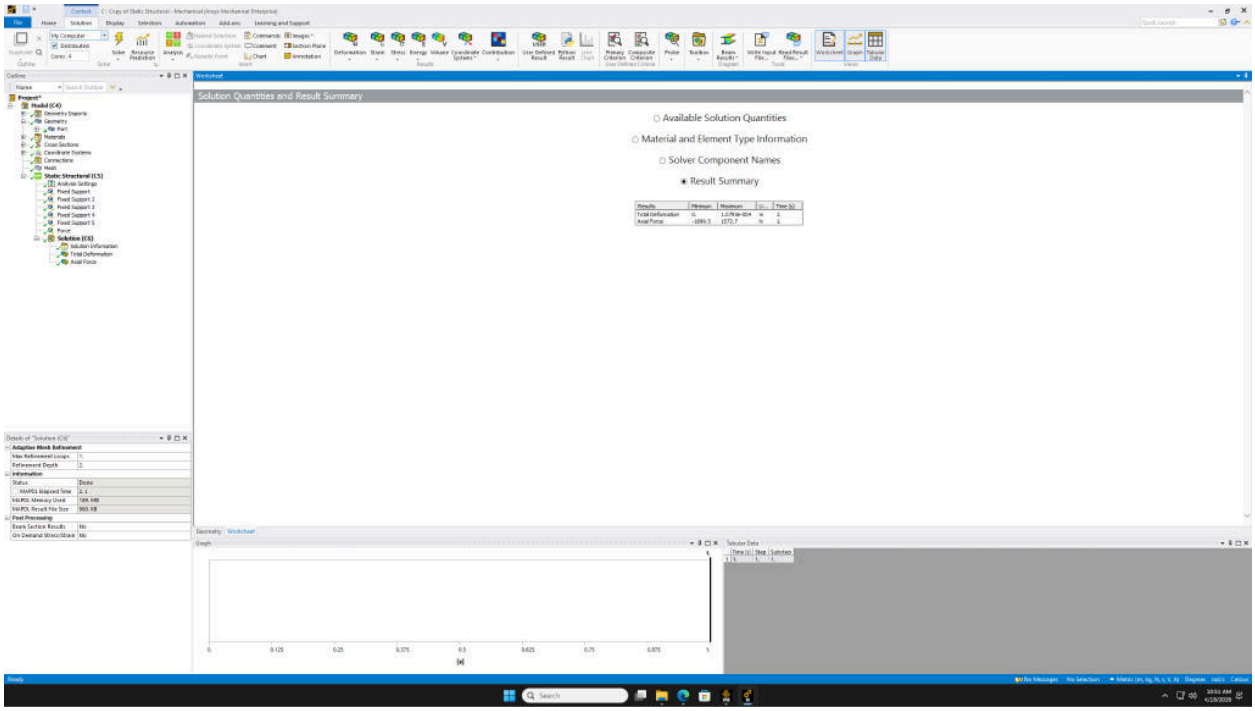
Case 3: 4cm diameter, not fixed





Case 2: 2 cm diameter, fixed





Case 4: 2cm diameter, not fixed

